

BLM2041 Signals and Systems

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BLM2041 Signals and Systems

Fourier Transform

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LECTURE OBJECTIVES

- Review
 - Frequency Response
 - Fourier Series
- Definition of Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

- Relation to Fourier Series
- Examples of Fourier transform pairs
- Basic properties of Fourier transforms
 - Convolution property
 - Multiplication property

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WHY use the Fourier transform?

- Manipulate the Frequency Spectrum
- Analog Communication Systems
 - AM: Amplitude Modulation; FM
 - What are the Building Blocks ?
 - Abstract Layer, not implementation
- Ideal Filters
 - mostly BPFs
- Frequency Shifters
 - aka Modulators, Mixers or Multipliers: $x(t) \times p(t)$

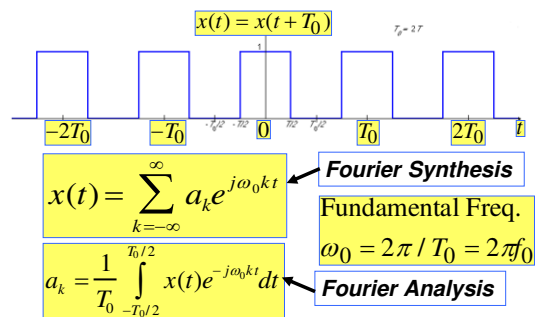
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Everything = Sum of Sinusoids

- One Square Pulse = Sum of Sinusoids
 - ??????????
- Finite Length
- Not Periodic
- Limit of Square Wave as Period $\rightarrow$ infinity
 - Intuitive Argument

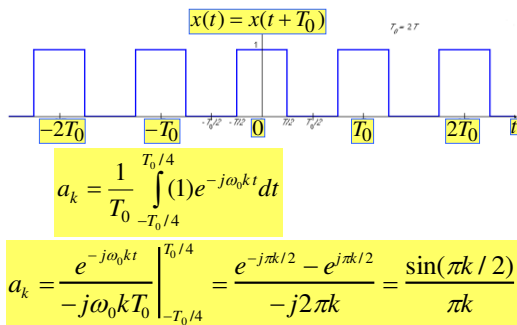
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Fourier Series: Periodic $x(t)$



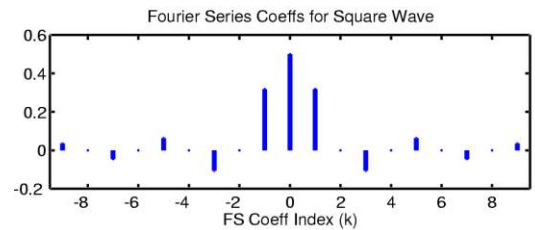
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Square Wave Signal



Spectrum from Fourier Series

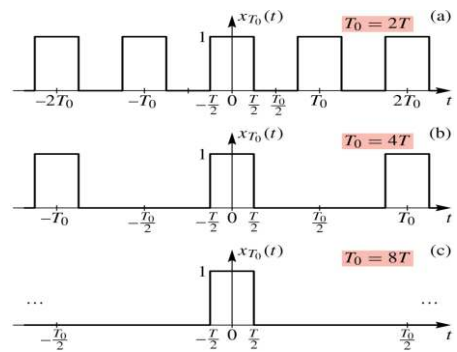
$$a_k = \frac{\sin(\pi k/2)}{\pi k} = \begin{cases} \neq 0 & k = 0, \pm 1, \pm 3, \dots \\ 0 & k = \pm 2, \pm 4, \dots \end{cases}$$



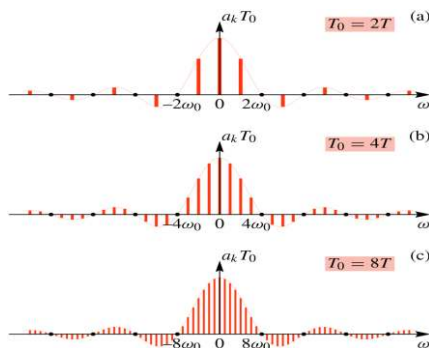
What if $x(t)$ is not periodic?

- Sum of Sinusoids?
 - Non-harmonically related sinusoids
 - Would not be periodic,
 - but would probably be non-zero for all t .
- Fourier transform
 - gives a "sum" (actually an **integral**) that involves **ALL** frequencies
 - can represent signals that are identically zero for negative t . !!!!!!!!!

Limiting Behavior of FS



Limiting Behavior of Spectrum



FS in the LIMIT (long period)

$$x_{T_0}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} (T_0 a_k) e^{j\omega_0 k t} \xrightarrow{\left(\frac{2\pi}{T_0}\right) \mapsto \omega} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

Fourier Synthesis

$$\lim_{T_0 \rightarrow \infty} \frac{2\pi}{T_0} = d\omega$$

$$\lim_{T_0 \rightarrow \infty} \frac{2\pi}{T_0} k = \omega$$

$$\lim_{T_0 \rightarrow \infty} T_0 a_k = X(j\omega)$$

$$T_0 a_k = \int_{-T_0/2}^{T_0/2} x_{T_0}(t) e^{-j\omega_0 k t} dt \mapsto X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Fourier Analysis

Fourier Transform Defined

- For non-periodic signals

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

Fourier Synthesis
(Inverse Transform)

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Fourier Analysis
(Forward Transform)

Time - Domain $\Leftrightarrow$ Frequency - Domain
 $x(t) \Leftrightarrow X(j\omega)$

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Example 1

$$x(t) = e^{-at} u(t)$$

$$X(j\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \int_0^{\infty} e^{-(a+j\omega)t} dt \quad a > 0$$

$$X(j\omega) = -\frac{e^{-at} e^{-j\omega t}}{a+j\omega} \Big|_0^{\infty} = \frac{1}{a+j\omega}$$

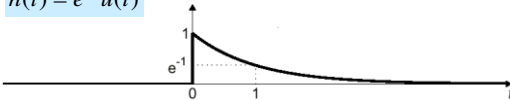
$$X(j\omega) = \frac{1}{a+j\omega}$$

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Example 1 - Frequency Response

- Fourier Transform of $h(t)$ is
 – the Frequency Response

$$h(t) = e^{-t} u(t)$$

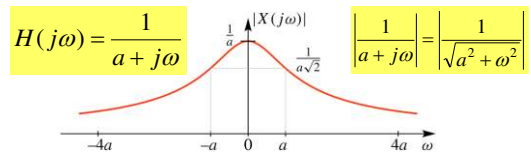


$$h(t) = e^{-t} u(t) \Leftrightarrow H(j\omega) = \frac{1}{1+j\omega}$$

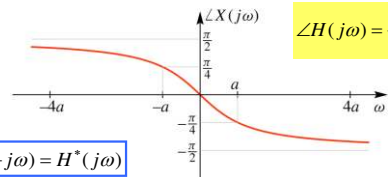
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Example 1 - Magnitude and Phase Plots

$$H(j\omega) = \frac{1}{a+j\omega}$$



$$\left| \frac{1}{a+j\omega} \right| = \frac{1}{\sqrt{a^2 + \omega^2}}$$



$$\angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$

$$H(-j\omega) = H^*(j\omega)$$

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Example 2

$$x(t) = \begin{cases} 1 & |t| < T/2 \\ 0 & |t| > T/2 \end{cases}$$

$$X(j\omega) = \int_{-T/2}^{T/2} (1) e^{-j\omega t} dt = \int_{-T/2}^{T/2} e^{-j\omega t} dt$$

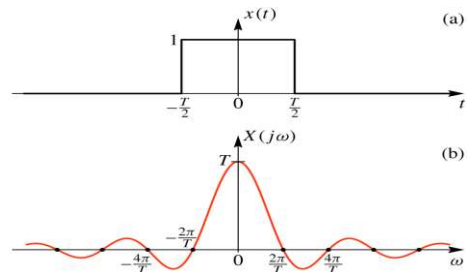
$$X(j\omega) = \frac{e^{-j\omega t}}{-j\omega} \Big|_{-T/2}^{T/2} = \frac{e^{-j\omega T/2} - e^{j\omega T/2}}{-j\omega}$$

$$X(j\omega) = \frac{\sin(\omega T/2)}{(\omega/2)}$$

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Example 2

$$x(t) = \begin{cases} 1 & |t| < T/2 \\ 0 & |t| > T/2 \end{cases} \Leftrightarrow X(j\omega) = \frac{\sin(\omega T/2)}{(\omega/2)}$$



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Example 3

$$X(j\omega) = \begin{cases} 1 & |\omega| < \omega_b \\ 0 & |\omega| > \omega_b \end{cases}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\omega_b}^{\omega_b} 1 e^{j\omega t} d\omega$$

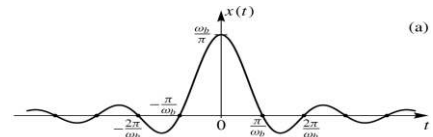
$$x(t) = \frac{1}{2\pi} \left. \frac{e^{j\omega t}}{jt} \right|_{-\omega_b}^{\omega_b} = \frac{1}{2\pi} \frac{e^{j\omega_b t} - e^{-j\omega_b t}}{jt}$$

$$x(t) = \frac{\sin(\omega_b t)}{\pi t}$$

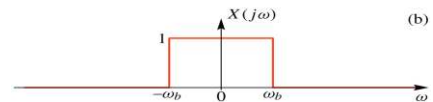
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Example 3

$$x(t) = \frac{\sin(\omega_b t)}{\pi t} \Leftrightarrow X(j\omega) = \begin{cases} 1 & |\omega| < \omega_b \\ 0 & |\omega| > \omega_b \end{cases}$$



(a)



(b)

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Example 4

$$x(t) = \delta(t - t_0)$$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

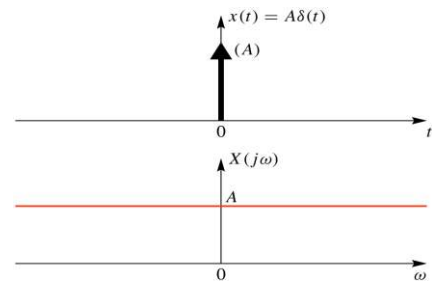
$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt = e^{-j\omega t_0}$$

Shifting Property of the Impulse

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Impulse function – Time and Frequency domains

$$x(t) = \delta(t) \Leftrightarrow X(j\omega) = 1$$



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Example 5

$$X(j\omega) = 2\pi\delta(\omega - \omega_0)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_0) e^{j\omega t} d\omega = e^{j\omega_0 t}$$

$$x(t) = e^{j\omega_0 t} \Leftrightarrow X(j\omega) = 2\pi\delta(\omega - \omega_0)$$

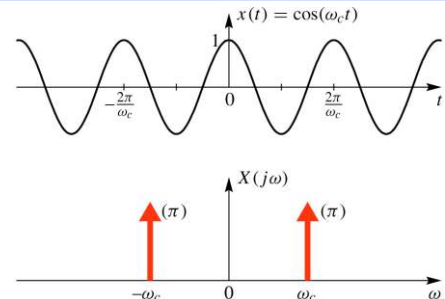
$$x(t) = 1 \Leftrightarrow X(j\omega) = 2\pi\delta(\omega)$$

$$x(t) = \cos(\omega_0 t) \Leftrightarrow X(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

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Example 5

$$x(t) = \cos(\omega_c t) \Leftrightarrow X(j\omega) = \pi\delta(\omega - \omega_c) + \pi\delta(\omega + \omega_c)$$



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Table of Fourier Transforms

$$x(t) = e^{-at} u(t) \Leftrightarrow X(j\omega) = \frac{1}{a + j\omega}$$

$$x(t) = \begin{cases} 1 & |t| < T/2 \\ 0 & |t| > T/2 \end{cases} \Leftrightarrow X(j\omega) = \frac{\sin(\omega T/2)}{(\omega/2)}$$

$$x(t) = \frac{\sin(\omega_0 t)}{(\pi t)} \Leftrightarrow X(j\omega) = \begin{cases} 1 & |\omega| < \omega_0 \\ 0 & |\omega| > \omega_0 \end{cases}$$

$$x(t) = \delta(t - t_0) \Leftrightarrow X(j\omega) = e^{-j\omega t_0}$$

$$x(t) = e^{j\omega_0 t} \Leftrightarrow X(j\omega) = 2\pi\delta(\omega - \omega_0)$$

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Fourier Transform of a General Periodic Signal

- If $x(t)$ is periodic with period T_0 ,

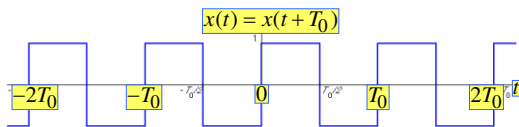
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jk\omega_0 t} dt$$

$$\text{Therefore, since } e^{jk\omega_0 t} \Leftrightarrow 2\pi\delta(\omega - k\omega_0)$$

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

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Square Wave Signal

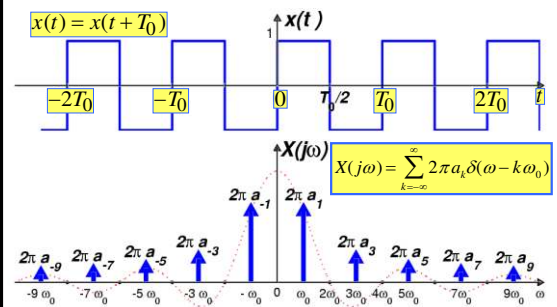


$$a_k = \frac{1}{T_0} \int_0^{T_0/2} (1) e^{-j\omega_0 k t} dt + \frac{1}{T_0} \int_{T_0/2}^{T_0} (-1) e^{-j\omega_0 k t} dt$$

$$a_k = \frac{e^{-j\omega_0 k t}}{-j\omega_0 k T_0} \Big|_0^{T_0/2} - \frac{e^{-j\omega_0 k t}}{-j\omega_0 k T_0} \Big|_{T_0/2}^{T_0} = \frac{1 - e^{-j\pi k}}{j\pi k}$$

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Square Wave Fourier Transform



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Table of Easy FT Properties

Linearity Property

$$ax_1(t) + bx_2(t) \Leftrightarrow aX_1(j\omega) + bX_2(j\omega)$$

Delay Property

$$x(t - t_d) \Leftrightarrow e^{-j\omega t_d} X(j\omega)$$

Frequency Shifting

$$x(t) e^{j\omega_0 t} \Leftrightarrow X(j(\omega - \omega_0))$$

Scaling

$$x(at) \Leftrightarrow \frac{1}{|a|} X(j(\frac{\omega}{a}))$$

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Scaling Property

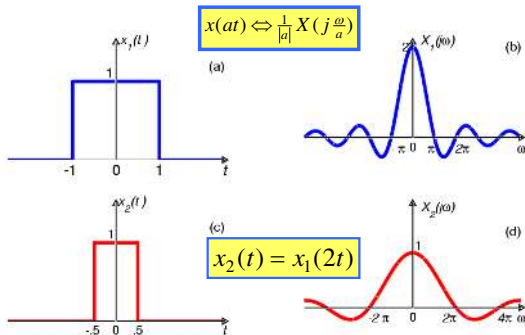
$$x(at) \Leftrightarrow \frac{1}{|a|} X(j\frac{\omega}{a})$$

$$\begin{aligned} \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt &= \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega(\lambda/a)} \frac{d\lambda}{|a|} \\ &= \frac{1}{|a|} X(j\frac{\omega}{a}) \end{aligned}$$

$$x(2t) \text{ shrinks; } \frac{1}{2} X(j\frac{\omega}{2}) \text{ expands}$$

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Scaling Property



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Uncertainty Principle

- Try to make $x(t)$ shorter
 - Then $X(j\omega)$ will get wider
 - Narrow pulses have wide bandwidth
- Try to make $X(j\omega)$ narrower
 - Then $x(t)$ will have longer duration
- **Cannot simultaneously reduce time duration and bandwidth**

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Significant FT Properties

$$x(t) * h(t) \Leftrightarrow H(j\omega)X(j\omega)$$

$$x(t)p(t) \Leftrightarrow \frac{1}{2\pi} X(j\omega) * P(j\omega)$$

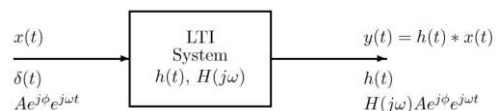
$$x(t)e^{j\omega_0 t} \Leftrightarrow X(j(\omega - \omega_0))$$

Differentiation Property

$$\frac{dx(t)}{dt} \Leftrightarrow (j\omega)X(j\omega)$$

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Convolution Property



- Convolution in the time-domain

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau$$

corresponds to **MULTIPLICATION** in the frequency-domain

$$Y(j\omega) = H(j\omega)X(j\omega)$$

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Convolution Example

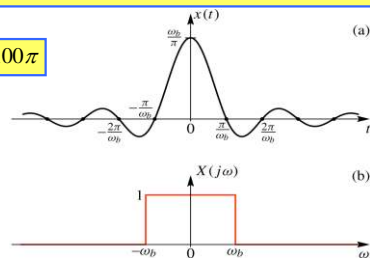
- Bandlimited **Input** Signal
 - “sinc” function
- Ideal LPF (Lowpass Filter)
 - $h(t)$ is a “sinc”
- **Output** is Bandlimited
 - Convolve “sincs”

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Ideally Bandlimited Signal

$$x(t) = \frac{\sin(100\pi t)}{\pi t} \Leftrightarrow X(j\omega) = \begin{cases} 1 & |\omega| < 100\pi \\ 0 & |\omega| > 100\pi \end{cases}$$

$$\omega_b = 100\pi$$

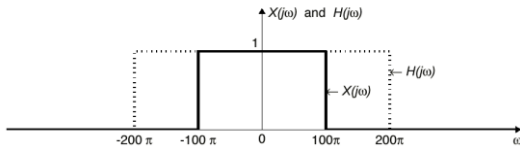


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Convolution Example 1

$$x(t) * h(t) \Leftrightarrow H(j\omega)X(j\omega)$$

$$\frac{\sin(100\pi t)}{\pi t} * \frac{\sin(200\pi t)}{\pi t} = \frac{\sin(100\pi t)}{\pi t}$$



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Cosine Input to LTI System

$$Y(j\omega) = H(j\omega)X(j\omega)$$

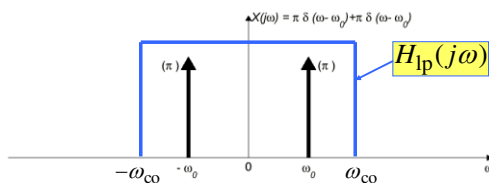
$$= H(j\omega)[\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)]$$

$$= H(j\omega_0)\pi\delta(\omega - \omega_0) + H(-j\omega_0)\pi\delta(\omega + \omega_0)$$

$$\begin{aligned} y(t) &= H(j\omega_0)\frac{1}{2}e^{j\omega_0 t} + H(-j\omega_0)\frac{1}{2}e^{-j\omega_0 t} \\ &= H(j\omega_0)\frac{1}{2}e^{j\omega_0 t} + H^*(j\omega_0)\frac{1}{2}e^{-j\omega_0 t} \\ &= |H(j\omega_0)|\cos(\omega_0 t + \angle H(j\omega_0)) \end{aligned}$$

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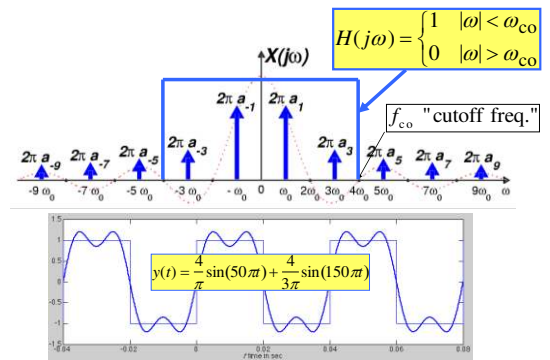
Ideal Lowpass Filter



$$\begin{aligned} y(t) &= x(t) & \text{if } \omega_0 < \omega_{co} \\ y(t) &= 0 & \text{if } \omega_0 > \omega_{co} \end{aligned}$$

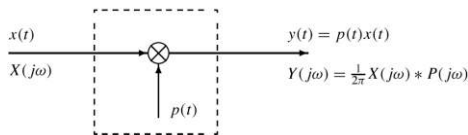
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Ideal Lowpass Filter



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Signal Multiplier (Modulator)



- Multiplication in the time-domain corresponds to convolution in the frequency-domain.

$$Y(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta)P(j(\omega - \theta))d\theta$$

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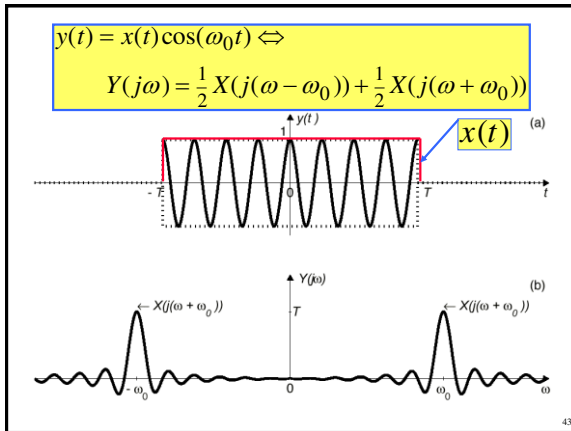
Frequency Shifting Property

$$x(t)e^{j\omega_0 t} \Leftrightarrow X(j(\omega - \omega_0))$$

$$\begin{aligned} \int_{-\infty}^{\infty} e^{j\omega_0 t} x(t) e^{-j\omega t} dt &= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt \\ &= X(j(\omega - \omega_0)) \end{aligned}$$

$$y(t) = \frac{\sin 7t}{\pi} e^{j\omega_0 t} \Leftrightarrow Y(j\omega) = \begin{cases} 1 & \omega_0 - 7 < \omega < \omega_0 + 7 \\ 0 & \text{elsewhere} \end{cases}$$

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Differentiation Property

$$\frac{dx(t)}{dt} = \frac{d}{dt} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \right)$$

Multiply by $j\omega$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega) X(j\omega) e^{j\omega t} d\omega$$

$$\frac{d}{dt} (e^{-at} u(t)) = -ae^{-at} u(t) + e^{-at} \delta(t)$$

$$= \delta(t) - ae^{-at} u(t)$$

$$\Leftrightarrow \frac{j\omega}{a + j\omega}$$

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